

**DR. D. Y. PATIL INSTITUTE OF ENGINEERING, MANAGEMENT AND RESEARCH**  
**Akurdi, Pune-411044**

**Department-Artificial Intelligence and Data Science**

**Laboratory Journal | Subject: Computer Laboratory – IV**

**AY:2025-26**

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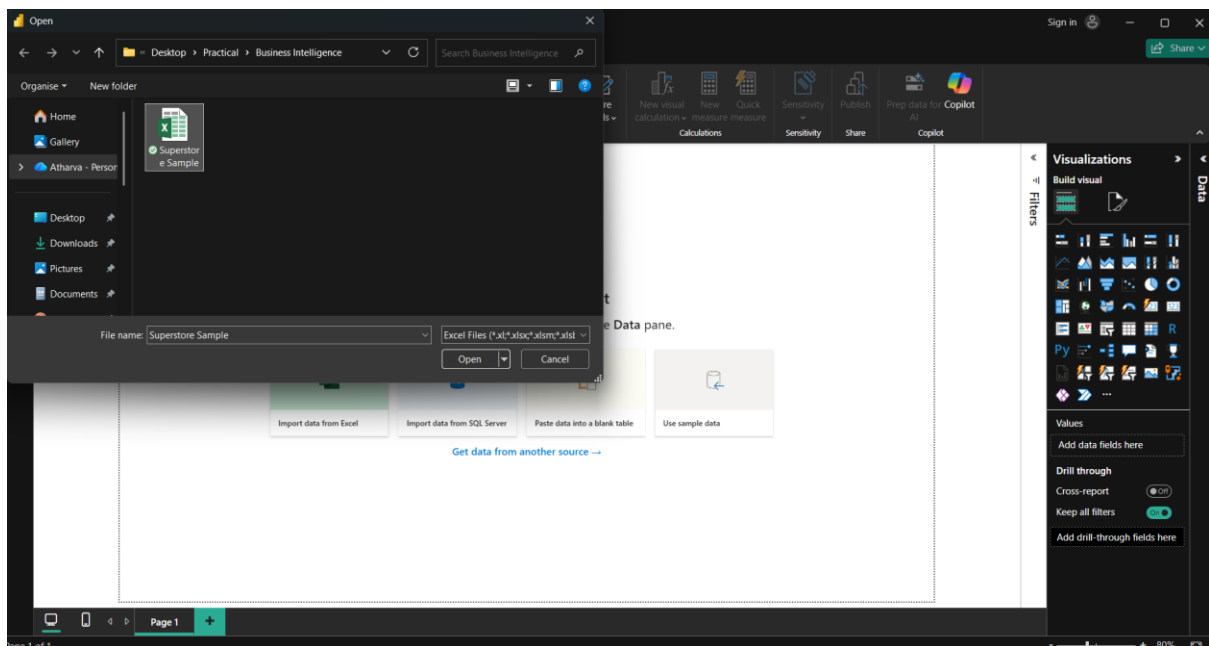
**Class: B.E. (A)**

**Roll No: 22508**

### Lab Practical

Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sql server / Power BI.

#### STEP 1: Extraction (Load Data)



Navigator

Display Options

- Superstore Sample.xlsx [1]
- Superstore

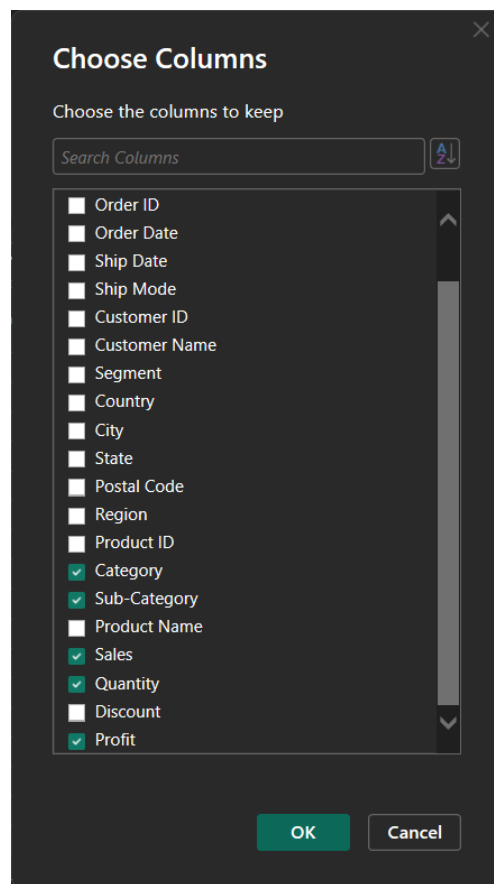
Superstore

| Row ID | Order ID       | Order Date | Ship Date  | Ship Mode      | Cu |
|--------|----------------|------------|------------|----------------|----|
| 1      | CA-2013-152156 | 09/11/2013 | 12/11/2013 | Second Class   | Cu |
| 2      | CA-2013-152156 | 09/11/2013 | 12/11/2013 | Second Class   | Cu |
| 3      | CA-2013-138688 | 13/06/2013 | 17/06/2013 | Second Class   | Cu |
| 4      | US-2012-108966 | 11/10/2012 | 18/10/2012 | Standard Class | Cu |
| 5      | US-2012-108966 | 11/10/2012 | 18/10/2012 | Standard Class | Cu |
| 6      | CA-2011-115812 | 09/06/2011 | 14/06/2011 | Standard Class | Cu |
| 7      | CA-2011-115812 | 09/06/2011 | 14/06/2011 | Standard Class | Cu |
| 8      | CA-2011-115812 | 09/06/2011 | 14/06/2011 | Standard Class | Cu |
| 9      | CA-2011-115812 | 09/06/2011 | 14/06/2011 | Standard Class | Cu |
| 10     | CA-2011-115812 | 09/06/2011 | 14/06/2011 | Standard Class | Cu |
| 11     | CA-2011-115812 | 09/06/2011 | 14/06/2011 | Standard Class | Cu |
| 12     | CA-2011-115812 | 09/06/2011 | 14/06/2011 | Standard Class | Cu |
| 13     | CA-2014-114412 | 16/04/2014 | 21/04/2014 | Standard Class | Cu |
| 14     | CA-2013-161389 | 06/12/2013 | 11/12/2013 | Standard Class | Cu |
| 15     | US-2012-118983 | 22/11/2012 | 26/11/2012 | Standard Class | Cu |
| 16     | US-2012-118983 | 22/11/2012 | 26/11/2012 | Standard Class | Cu |
| 17     | CA-2011-105893 | 11/11/2011 | 18/11/2011 | Standard Class | Cu |
| 18     | CA-2011-167164 | 13/05/2011 | 15/05/2011 | Second Class   | Cu |
| 19     | CA-2011-143336 | 27/08/2011 | 01/09/2011 | Second Class   | Cu |
| 20     | CA-2011-143336 | 27/08/2011 | 01/09/2011 | Second Class   | Cu |
| 21     | CA-2011-143336 | 27/08/2011 | 01/09/2011 | Second Class   | Cu |
| 22     | CA-2013-137330 | 10/12/2013 | 14/12/2013 | Standard Class | Cu |
| 23     | CA-2013-137330 | 10/12/2013 | 14/12/2013 | Standard Class | Cu |

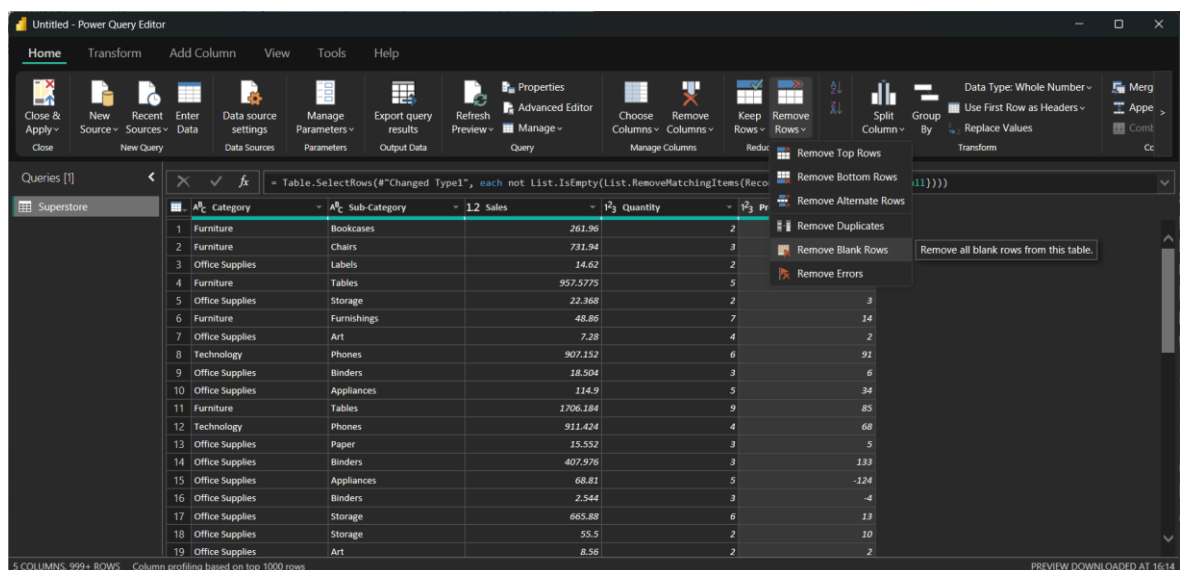
Load Transform Data Cancel

## STEP 2: Transformation

### (A) Remove Unnecessary Columns (Select the ones you want to keep)



### (B) Handle Missing Values



## (C) Changing the Data Type:

The screenshot shows the Power Query Editor interface. The 'Data Type' dropdown menu is open for the 'Sales' column, showing options like 'Decimal Number', 'Fixed decimal number', 'Whole Number', 'Percentage', 'Date/Time', 'Date', 'Time', 'Date/Time/Timezone', 'Duration', 'Text', 'True/False', and 'Binary'. The 'Sales' column is highlighted in the table.

| Category        | Sub-Category | Sales    | Quantity | Profit   |
|-----------------|--------------|----------|----------|----------|
| Furniture       | Bookcases    | 261.96   | 2        | 41.9136  |
| Furniture       | Chairs       | 731.94   | 3        | 219.582  |
| Office Supplies | Labels       | 14.62    | 2        | 6.8714   |
| Furniture       | Tables       | 957.5775 | 5        | -383.031 |
| Office Supplies | Storage      | 22.368   | 2        | 2.5164   |
| Furniture       | Furnishings  | 48.86    | 7        | 14.1694  |
| Office Supplies | Art          | 7.28     | 4        | 1.9656   |
| Technology      | Phones       | 907.152  | 6        | 90.7152  |
| Office Supplies | Binders      | 18.504   | 3        | 5.7825   |
| Office Supplies | Appliances   | 114.9    | 5        | 34.47    |
| Furniture       | Tables       | 1706.184 | 9        | 85.3092  |
| Technology      | Phones       | 911.424  | 4        | 68.3568  |
| Office Supplies | Paper        | 15.552   | 3        | 5.4432   |
| Office Supplies | Binders      | 407.976  | 3        | 132.5922 |
| Office Supplies | Appliances   | 68.81    | 5        | -123.858 |
| Office Supplies | Binders      | 2.544    | 3        | -3.816   |
| Office Supplies | Storage      | 665.88   | 6        | 13.3176  |
| Office Supplies | Storage      | 55.5     | 2        | 9.99     |
| Office Supplies | Art          | 8.56     | 2        | 2.4824   |

## (D) Create NEW Column

The screenshot shows the Power Query Editor interface with the 'Add Column' tab selected. The 'Custom Column' formula bar contains the formula: `= Table.SelectRows(#"Changed Type1", each not List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues(_), {"", null})))`. The table below shows the data after applying the formula.

| Category        | Sub-Category | Sales    | Quantity | Profit |
|-----------------|--------------|----------|----------|--------|
| Furniture       | Bookcases    | 261.96   | 2        | 42     |
| Furniture       | Chairs       | 731.94   | 3        | 220    |
| Office Supplies | Labels       | 14.62    | 2        | 7      |
| Furniture       | Tables       | 957.5775 | 5        | -383   |
| Office Supplies | Storage      | 22.368   | 2        | 3      |
| Furniture       | Furnishings  | 48.86    | 7        | 14     |
| Office Supplies | Art          | 7.28     | 4        | 2      |
| Technology      | Phones       | 907.152  | 6        | 91     |
| Office Supplies | Binders      | 18.504   | 3        | 6      |
| Office Supplies | Appliances   | 114.9    | 5        | 34     |
| Furniture       | Tables       | 1706.184 | 9        | 85     |
| Technology      | Phones       | 911.424  | 4        | 68     |
| Office Supplies | Paper        | 15.552   | 3        | 5      |
| Office Supplies | Binders      | 407.976  | 3        | 133    |
| Office Supplies | Appliances   | 68.81    | 5        | -124   |
| Office Supplies | Binders      | 2.544    | 3        | -4     |
| Office Supplies | Storage      | 665.88   | 6        | 13     |
| Office Supplies | Storage      | 55.5     | 2        | 10     |
| Office Supplies | Art          | 8.56     | 2        | 2      |

The 'Custom Column' dialog box is shown. It has a 'New column name' field with 'Profit Margin' entered. The 'Custom column formula' field contains the formula: `= [Profit] / [Sales] * 100`. The 'Available columns' list on the right includes 'Category', 'Sub-Category', 'Sales', 'Quantity', and 'Profit'. The 'OK' button is highlighted.

Custom Column

Add a column that is computed from the other columns.

New column name

Profit Margin

Custom column formula

`= [Profit] / [Sales] * 100`

Available columns

- Category
- Sub-Category
- Sales
- Quantity
- Profit

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

## STEP 3: Loading

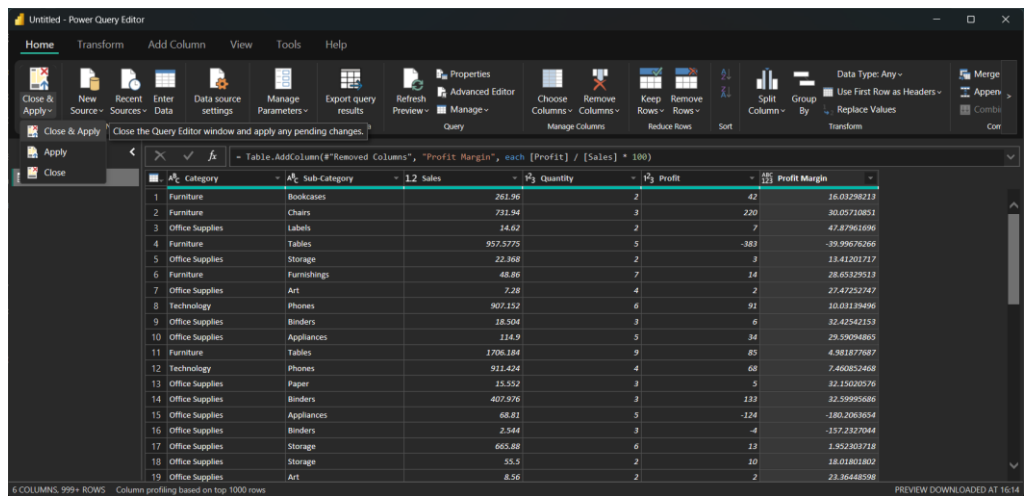


Table: AddColumn(8"Removed Column", "Profit Margin", each [Profit] / [Sales] \* 100)

|    | Category        | Sub-Category | Sales    | Quantity | Profit | Profit Margin |
|----|-----------------|--------------|----------|----------|--------|---------------|
| 1  | Furniture       | Bookcases    | 261.96   | 2        | 42     | 16.03298213   |
| 2  | Furniture       | Chairs       | 731.94   | 3        | 220    | 30.05710851   |
| 3  | Office Supplies | Labels       | 14.62    | 2        | 7      | 47.87961696   |
| 4  | Furniture       | Tables       | 957.5775 | 5        | -383   | -39.99676266  |
| 5  | Office Supplies | Storage      | 22.368   | 2        | 3      | 11.41201717   |
| 6  | Furniture       | Furnishings  | 48.86    | 7        | 14     | 28.65329513   |
| 7  | Office Supplies | Art          | 7.28     | 4        | 2      | 27.47251747   |
| 8  | Technology      | Phones       | 807.152  | 6        | 91     | 10.03139496   |
| 9  | Office Supplies | Binders      | 16.504   | 3        | 6      | 32.42421151   |
| 10 | Office Supplies | Appliances   | 114.9    | 5        | 38     | 29.59094865   |
| 11 | Furniture       | Tables       | 1706.184 | 9        | 83     | 4.881877687   |
| 12 | Technology      | Phones       | 911.424  | 4        | 68     | 7.460252468   |
| 13 | Office Supplies | Paper        | 15.552   | 3        | 5      | 32.15020576   |
| 14 | Office Supplies | Binders      | 407.976  | 3        | 133    | 32.59995686   |
| 15 | Office Supplies | Appliances   | 68.81    | 5        | -124   | -180.2063654  |
| 16 | Office Supplies | Binders      | 2.544    | 3        | -4     | -157.2327044  |
| 17 | Office Supplies | Storage      | 665.88   | 6        | 13     | 1.952303718   |
| 18 | Office Supplies | Storage      | 55.3     | 2        | 10     | 18.01801802   |
| 19 | Office Supplies | Art          | 8.56     | 2        | 2      | 23.36448598   |

6 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 16:14

## STEP 4: Database Construction

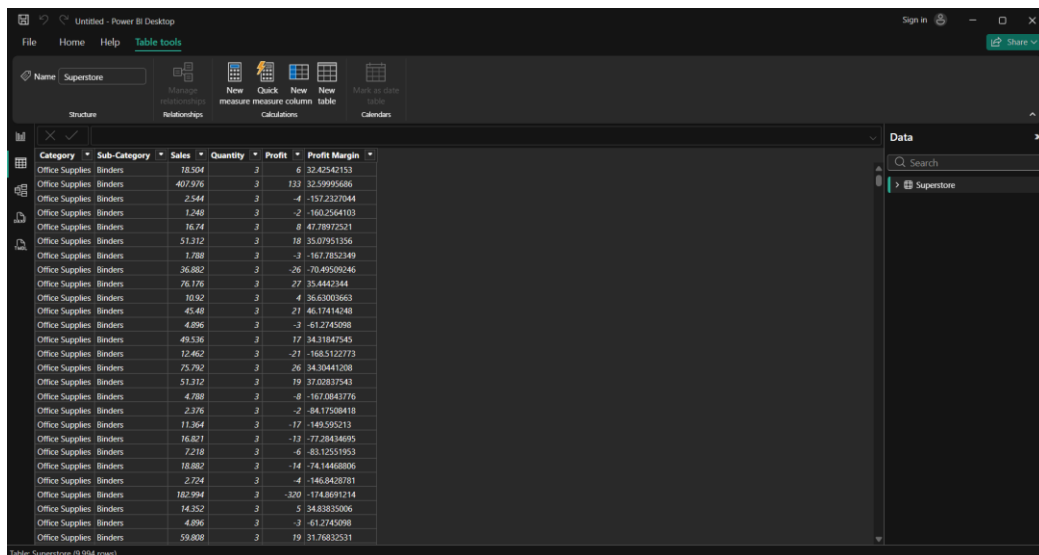


Table: Superstore (9,994 rows)

| Category        | Sub-Category | Sales   | Quantity | Profit | Profit Margin |
|-----------------|--------------|---------|----------|--------|---------------|
| Office Supplies | Binders      | 18.504  | 3        | 6      | 32.42421151   |
| Office Supplies | Binders      | 407.976 | 3        | 133    | 32.59995686   |
| Office Supplies | Binders      | 2.544   | 3        | -4     | -157.2327044  |
| Office Supplies | Binders      | 1.248   | 3        | -2     | -160.2564103  |
| Office Supplies | Binders      | 16.74   | 3        | 8      | 47.78972521   |
| Office Supplies | Binders      | 51.312  | 3        | 18     | 35.07951356   |
| Office Supplies | Binders      | 1.788   | 3        | -3     | -167.7852349  |
| Office Supplies | Binders      | 36.882  | 3        | -26    | -70.49509246  |
| Office Supplies | Binders      | 26.176  | 3        | 27     | 35.4442144    |
| Office Supplies | Binders      | 10.82   | 3        | 4      | 36.63020662   |
| Office Supplies | Binders      | 45.48   | 3        | 21     | 46.17414248   |
| Office Supplies | Binders      | 4.896   | 3        | -3     | -61.2745098   |
| Office Supplies | Binders      | 49.536  | 3        | 17     | 34.31847545   |
| Office Supplies | Binders      | 12.462  | 3        | -21    | -168.5122773  |
| Office Supplies | Binders      | 75.792  | 3        | 26     | 34.30441208   |
| Office Supplies | Binders      | 51.312  | 3        | 19     | 37.02837543   |
| Office Supplies | Binders      | 4.788   | 3        | -8     | -162.0443776  |
| Office Supplies | Binders      | 2.316   | 3        | -2     | -84.17508418  |
| Office Supplies | Binders      | 11.364  | 3        | -17    | -149.595213   |
| Office Supplies | Binders      | 16.821  | 3        | -13    | -77.28434695  |
| Office Supplies | Binders      | 7.218   | 3        | -6     | -83.12551953  |
| Office Supplies | Binders      | 18.882  | 3        | -14    | -74.14468806  |
| Office Supplies | Binders      | 2.724   | 3        | -4     | -146.8426781  |
| Office Supplies | Binders      | 182.994 | 3        | -320   | -174.8691214  |
| Office Supplies | Binders      | 14.252  | 3        | 5      | 34.83838206   |
| Office Supplies | Binders      | 4.896   | 3        | -3     | -61.2745098   |
| Office Supplies | Binders      | 59.808  | 3        | 19     | 31.76832531   |

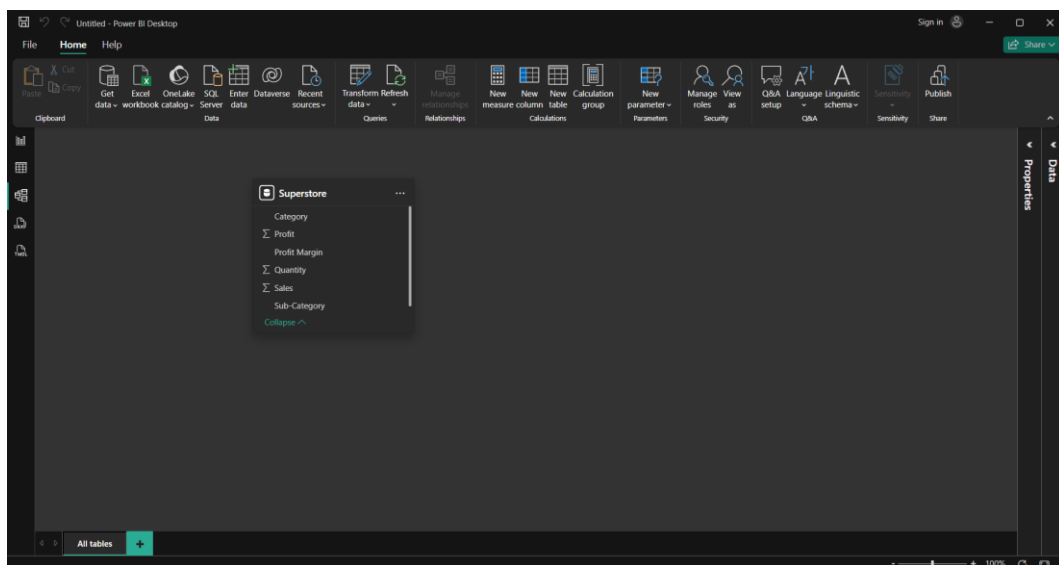


Table: Superstore (9,994 rows)

| Category        | Sub-Category | Sales   | Quantity | Profit | Profit Margin |
|-----------------|--------------|---------|----------|--------|---------------|
| Office Supplies | Binders      | 18.504  | 3        | 6      | 32.42421151   |
| Office Supplies | Binders      | 407.976 | 3        | 133    | 32.59995686   |
| Office Supplies | Binders      | 2.544   | 3        | -4     | -157.2327044  |
| Office Supplies | Binders      | 1.248   | 3        | -2     | -160.2564103  |
| Office Supplies | Binders      | 16.74   | 3        | 8      | 47.78972521   |
| Office Supplies | Binders      | 51.312  | 3        | 18     | 35.07951356   |
| Office Supplies | Binders      | 1.788   | 3        | -3     | -167.7852349  |
| Office Supplies | Binders      | 36.882  | 3        | -26    | -70.49509246  |
| Office Supplies | Binders      | 26.176  | 3        | 27     | 35.4442144    |
| Office Supplies | Binders      | 10.82   | 3        | 4      | 36.63020662   |
| Office Supplies | Binders      | 45.48   | 3        | 21     | 46.17414248   |
| Office Supplies | Binders      | 4.896   | 3        | -3     | -61.2745098   |
| Office Supplies | Binders      | 49.536  | 3        | 17     | 34.31847545   |
| Office Supplies | Binders      | 12.462  | 3        | -21    | -168.5122773  |
| Office Supplies | Binders      | 75.792  | 3        | 26     | 34.30441208   |
| Office Supplies | Binders      | 51.312  | 3        | 19     | 37.02837543   |
| Office Supplies | Binders      | 4.788   | 3        | -8     | -162.0443776  |
| Office Supplies | Binders      | 2.316   | 3        | -2     | -84.17508418  |
| Office Supplies | Binders      | 11.364  | 3        | -17    | -149.595213   |
| Office Supplies | Binders      | 16.821  | 3        | -13    | -77.28434695  |
| Office Supplies | Binders      | 7.218   | 3        | -6     | -83.12551953  |
| Office Supplies | Binders      | 18.882  | 3        | -14    | -74.14468806  |
| Office Supplies | Binders      | 2.724   | 3        | -4     | -146.8426781  |
| Office Supplies | Binders      | 182.994 | 3        | -320   | -174.8691214  |
| Office Supplies | Binders      | 14.252  | 3        | 5      | 34.83838206   |
| Office Supplies | Binders      | 4.896   | 3        | -3     | -61.2745098   |
| Office Supplies | Binders      | 59.808  | 3        | 19     | 31.76832531   |

Sign: \_\_\_\_\_.